

ZHANGYE, China

WMO: 526520

Lat: 39.0795N

Lon: 100.2750E

Elev: 4797

StdP: 12.32

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.0	(c) 0.1	(d) -15.0	(e) 2.9	(f) 8.4	(g) -11.6	(h) 3.5	(i) 11.8	(j) 17.3	(k) 23.1	(l) 14.5	(m) 24.6	(n) 3.3	(o) 200	(p) 0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	92.2	62.7	89.1	61.4	86.4	60.4	67.1	82.1	65.2	80.5	63.8	78.7	5.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	102.7	71.6	60.9	95.6	69.8	59.2	89.7	69.2	34.7	82.1	33.0	80.2	31.8	78.8	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	14.2	11.9	-9.0	97.8	4.9	2.6	-12.5	99.7	-15.4	101.2	-18.2	102.6	-21.8	104.5
			-9.6	69.8	4.8	2.5	-13.0	71.6	-15.8	73.1	-18.5	74.5	-22.0	76.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	17.0	25.7	38.7	52.6	62.2	70.1	73.8	71.3	61.8	48.3	33.8	20.5		
	DBStd	20.75	7.49	7.92	8.12	7.91	6.47	5.24	5.22	5.14	5.66	6.59	7.09	7.32		
	HDD50	3637	1024	679	357	63	3	0	0	0	1	111	486	913		
	HDD65	6890	1489	1099	815	376	128	16	2	9	125	517	936	1378		
	CDD50	2943	0	0	7	140	381	603	738	659	356	59	0	0		
	CDD65	722	0	0	0	3	41	170	275	203	30	0	0	0		
	CDH74	7812	0	0	1	103	634	1834	2854	1990	378	18	0	0		
	CDH80	2862	0	0	0	20	152	633	1247	732	76	2	0	0		
(14) Wind	WSAvg	5.3	4.5	5.3	6.0	6.4	6.1	5.6	5.4	5.1	4.8	4.6	4.7	4.6		
(15) Precipitation	PrecAvg	8.2	0.1	0.1	0.2	0.4	0.9	1.4	1.9	1.6	1.1	0.4	0.1	0.0		
	PrecMax	10.1	0.3	0.2	0.7	0.7	1.4	2.5	3.0	3.0	2.0	0.7	0.3	0.1		
	PrecMin	6.2	0.0	0.0	0.0	0.2	0.3	0.6	1.1	0.8	0.2	0.1	0.0	0.0		
	PrecStd	1.0	0.0	0.0	0.2	0.2	0.3	0.5	0.5	0.5	0.4	0.2	0.1	0.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.3	56.2	70.1	82.7	88.2	93.2	97.4	94.5	86.5	76.4	62.5	49.5		
		MCWB	33.1	37.2	44.3	51.0	55.9	60.8	65.4	64.7	58.8	50.7	42.1	34.5		
	2%	DB	40.9	49.7	63.7	76.6	84.2	89.2	93.5	90.1	81.8	71.4	57.3	42.9		
		MCWB	28.9	34.1	41.8	48.9	54.4	59.1	63.9	62.6	56.4	49.3	40.0	30.8		
	5%	DB	35.3	45.0	59.1	72.2	80.7	86.4	90.2	87.2	78.3	67.1	52.9	38.1		
		MCWB	25.8	31.5	39.8	47.3	52.9	58.5	63.0	61.6	55.7	47.5	37.8	28.2		
	10%	DB	30.4	40.8	54.8	68.5	77.3	83.8	87.1	84.3	74.9	63.0	48.2	33.8		
		MCWB	22.7	29.2	38.0	45.9	51.8	58.0	62.3	61.1	54.9	45.8	35.7	25.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.4	37.8	45.2	53.1	58.6	65.7	71.2	69.3	63.6	53.6	43.1	34.6		
		MCDB	48.5	54.9	66.9	77.1	82.4	82.4	89.0	82.8	73.4	68.0	60.3	48.6		
	2%	WB	29.0	34.3	42.6	50.3	56.2	63.4	67.7	66.9	60.4	50.8	40.4	31.0		
		MCDB	40.1	48.9	61.1	72.6	78.1	79.2	83.5	81.0	72.4	66.7	55.8	42.3		
	5%	WB	25.9	31.9	40.6	48.4	54.6	61.9	65.8	64.9	58.8	48.9	38.3	28.4		
		MCDB	34.9	43.9	57.8	69.1	75.3	77.7	81.6	79.5	71.0	64.1	52.0	37.6		
	10%	WB	23.0	29.6	38.4	46.6	53.1	60.4	64.4	63.3	57.2	47.0	36.0	25.9		
		MCDB	30.1	40.1	53.9	66.0	72.8	76.7	79.8	77.5	69.0	61.0	47.8	33.4		
(35) Mean Daily Temperature Range	MDBR		25.5	26.0	25.1	23.9	23.6	22.8	21.8	22.1	23.2	26.3	24.8	24.2		
	5% DB	MCDBR	32.5	32.4	32.2	30.1	30.1	28.4	28.0	27.8	29.2	31.7	31.9	30.2		
		MCWBR	22.4	19.9	16.0	12.3	10.8	9.6	9.2	8.9	9.3	13.2	17.0	20.1		
	5% WB	MCDBR	31.6	30.5	30.5	26.8	26.1	23.0	22.9	21.7	21.2	28.4	30.4	29.1		
		MCWBR	21.9	18.7	15.2	10.9	10.1	9.1	8.7	8.4	8.2	11.9	16.4	19.5		
(40) Clear-Sky Solar Irradiance	taub		0.310	0.363	0.435	0.486	0.492	0.490	0.476	0.470	0.440	0.395	0.342	0.307		
	taud		2.201	2.059	1.887	1.796	1.847	1.924	2.034	2.037	2.050	2.071	2.142	2.215		
	Ebn at Noon		268	266	257	252	253	254	256	254	253	251	251	259		
	Edh at Noon		32	43	57	67	65	60	53	52	48	42	34	30		
(44) All-Sky Solar Radiation	RadAvg		882	1187	1569	1893	2089	2077	1996	1813	1554	1324	999	806		
	RadStd		55	49	96	102	85	126	127	126	101	69	47	52		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+261	+113
(47) Regional (3 neighbors)		+0.63	N/A	N/A	N/A	N/A	N/A	N/A	-178	+119	N/A

Nomenclature: See separate page